

EXP NO: 5	CLUSTERING WITH K-MEANS AND DIMENSIONALITY REDUCTION WITH PCA
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AIM

To implement **K-Means clustering** for grouping data points and **PCA** for dimensionality reduction and visualization using a suitable dataset.

ALGORITHM

1. K-Means Clustering

1. Generate or load the dataset.
2. Apply the **Elbow Method** to determine the optimal number of clusters.
3. Initialize the K-Means model with the selected number of clusters.
4. Assign data points to the nearest centroid and update centroids iteratively.
5. Visualize the clusters and calculate the **Silhouette Score**.

2. Principal Component Analysis (PCA)

1. Generate or load a multi-dimensional dataset.
2. Standardize the features using **StandardScaler**.
3. Apply PCA to reduce the dimensions to two principal components.
4. Calculate the explained variance of the components.
5. Visualize the reduced data and apply K-Means clustering on the PCA data.

CODE:

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.datasets import make_blobs
```

```

from sklearn.preprocessing import StandardScaler

from sklearn.cluster import KMeans

from sklearn.decomposition import PCA

from sklearn.metrics import silhouette_score

print("--- Part 1: K-Means Clustering ---")

X, y = make_blobs(

    n_samples=300,

    centers=3,

    cluster_std=0.60,

    random_state=42

)

df_kmeans = pd.DataFrame(

    X,

    columns=['Feature_1', 'Feature_2']

)

print("\nOriginal K-Means Dataset Head:")

print(df_kmeans.head())

```

```

wcss = []

for i in range(1, 11):

    kmeans = KMeans(

        n_clusters=i,

        init='k-means++',

```

```
        max_iter=300,
        n_init=10,
        random_state=42
    )
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(10, 6))
plt.plot(
    range(1, 11),
    wcss,
    marker='o',
    linestyle='--'
)

plt.title('Elbow Method for Optimal K (K-Means)')
plt.xlabel('Number of Clusters (K)')
plt.ylabel('WCSS')
plt.grid(True)
plt.show()

optimal_k = 3

kmeans = KMeans(
    n_clusters=optimal_k,
    init='k-means++',
    max_iter=300,
    n_init=10,
```

```
    random_state=42
)
clusters = kmeans.fit_predict(X)
df_kmeans['Cluster'] = clusters
plt.figure(figsize=(10, 8))
sns.scatterplot(
    x='Feature_1',
    y='Feature_2',
    hue='Cluster',
    data=df_kmeans,
    palette='viridis',
    s=100,
    alpha=0.8
)
plt.scatter(
    kmeans.cluster_centers_[0],
    kmeans.cluster_centers_[1],
    s=300,
    c='red',
    marker='X',
    label='Centroids'
)
plt.title(f'K-Means Clustering with K={optimal_k}')
plt.xlabel('Feature 1')
```

```

plt.ylabel('Feature 2')

plt.legend()

plt.grid(True)

plt.show()

silhouette_avg = silhouette_score(X, clusters)

print(
    f"\nSilhouette Score for K-Means (K={optimal_k}): "
    f"{silhouette_avg:.3f}"
)

print("\n--- Part 2: Dimensionality Reduction with PCA ---")

X_pca, y_pca = make_blobs(
    n_samples=500,
    n_features=4,
    centers=4,
    cluster_std=1.0,
    random_state=25
)

df_pca_original = pd.DataFrame(
    X_pca,
    columns=[f'Feature_{i+1}' for i in range(X_pca.shape[1])]
)

df_pca_original['True_Cluster'] = y_pca

print("\nOriginal PCA Dataset Head:")

print(df_pca_original.head())

```

```
print(
    f"Original PCA Dataset Shape: "
    f"{df_pca_original.shape}"
)

scaler = StandardScaler()

X_pca_scaled = scaler.fit_transform(X_pca)

pca = PCA(n_components=2)

principal_components = pca.fit_transform(X_pca_scaled)

df_principal_components = pd.DataFrame(
    principal_components,
    columns=[
        'Principal_Component_1',
        'Principal_Component_2'
    ]
)

df_principal_components['True_Cluster'] = y_pca

explained_variance = pca.explained_variance_ratio_

print("\nPrincipal Components Head:")

print(df_principal_components.head())

print(
    f"\nExplained Variance Ratio: "
    f"{explained_variance}"
)

print(
```

```
f'Total Explained Variance by 2 PCs: "  
f'{explained_variance.sum():.3f}' "  
)  
plt.figure(figsize=(10, 8))  
sns.scatterplot(  
    x='Principal_Component_1',  
    y='Principal_Component_2',  
    hue='True_Cluster',  
    data=df_principal_components,  
    palette='Paired',  
    s=100,  
    alpha=0.8  
)  
plt.title('PCA - Dimensionality Reduction to 2 Components')  
plt.xlabel(  
    f'PC1 ({explained_variance[0] * 100:.2f}%)'  
)  
plt.ylabel(  
    f'PC2 ({explained_variance[1] * 100:.2f}%)'  
)  
plt.grid(True)  
plt.show()  
kmeans_pca = KMeans(  
    n_clusters=4,
```

```

init='k-means++',
max_iter=300,
n_init=10,
random_state=42
)
clusters_pca = kmeans_pca.fit_predict(
    principal_components
)
df_principal_components[
    'KMeans_Cluster_on_PCA'
] = clusters_pca
plt.figure(figsize=(10, 8))
sns.scatterplot(
    x='Principal_Component_1',
    y='Principal_Component_2',
    hue='KMeans_Cluster_on_PCA',
    data=df_principal_components,
    palette='viridis',
    s=100,
    alpha=0.8
)
plt.scatter(
    kmeans_pca.cluster_centers[:, 0],
    kmeans_pca.cluster_centers[:, 1],

```

```

s=300,

c='red',

marker='X',

label='Centroids'

)

plt.title('K-Means Clustering on PCA-Reduced Data')

plt.xlabel('Principal Component 1')

plt.ylabel('Principal Component 2')

plt.legend()

plt.grid(True)

plt.show()

silhouette_avg_pca = silhouette_score(

    principal_components,

    clusters_pca

)

print(

    f"\nSilhouette Score for K-Means on PCA "

    f"(K=4): {silhouette_avg_pca:.3f}"

)

```

OUTPUT:

--- Part 1: K-Means Clustering ---

Original K-Means Dataset Head:

```

Feature_1 Feature_2
0 -7.155244 -7.390016

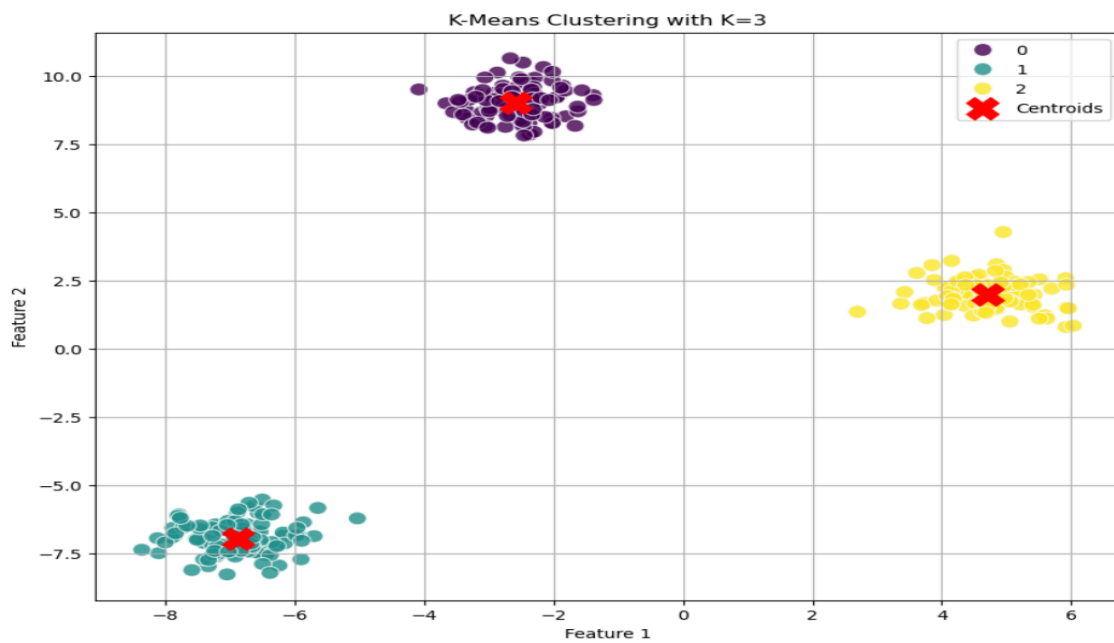
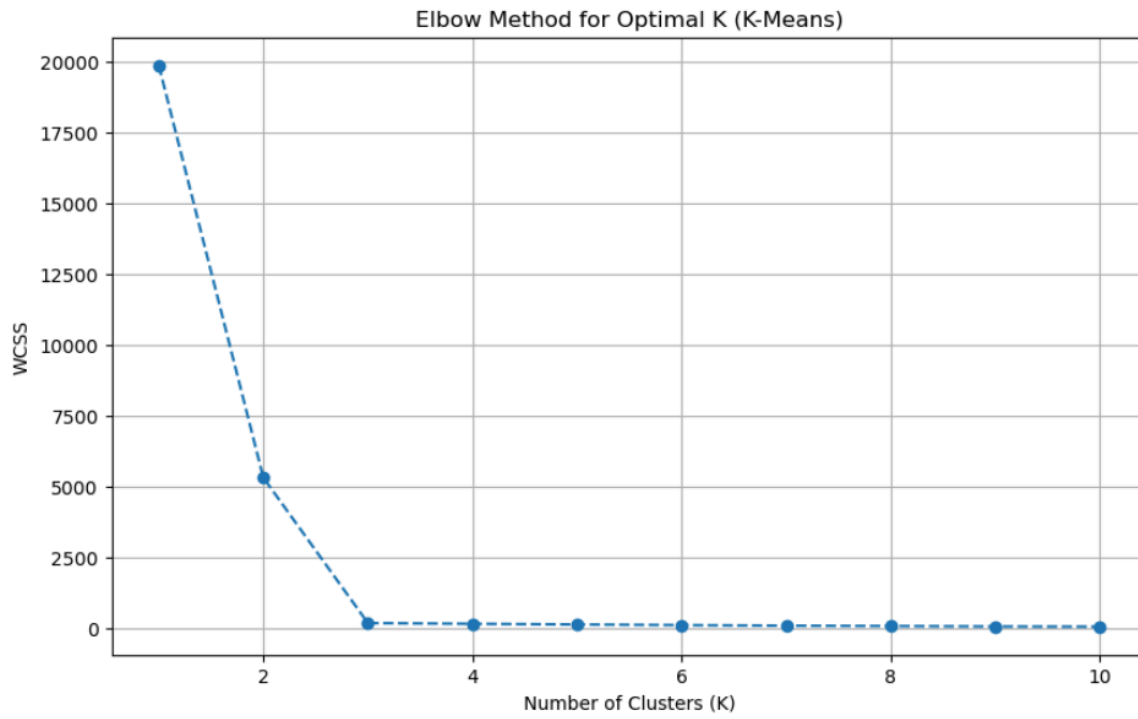
```

1 -7.395875 -7.110843

2 -2.015671 8.281780

3 4.509270 2.632436

4 -8.102502 -7.484961



Silhouette Score for K-Means (K=3): 0.908

--- Part 2: Dimensionality Reduction with PCA ---

Original PCA Dataset Head:

	Feature_1	Feature_2	Feature_3	Feature_4	True_Cluster
0	-0.638667	1.110057	-6.400722	-0.204990	3
1	-2.951556	-7.657445	3.844794	0.903589	1
2	-0.253177	2.125103	-7.869801	0.559678	3
3	-2.151209	3.401400	-5.734930	0.965230	3
4	-2.347519	-7.230467	3.478891	-0.443440	1

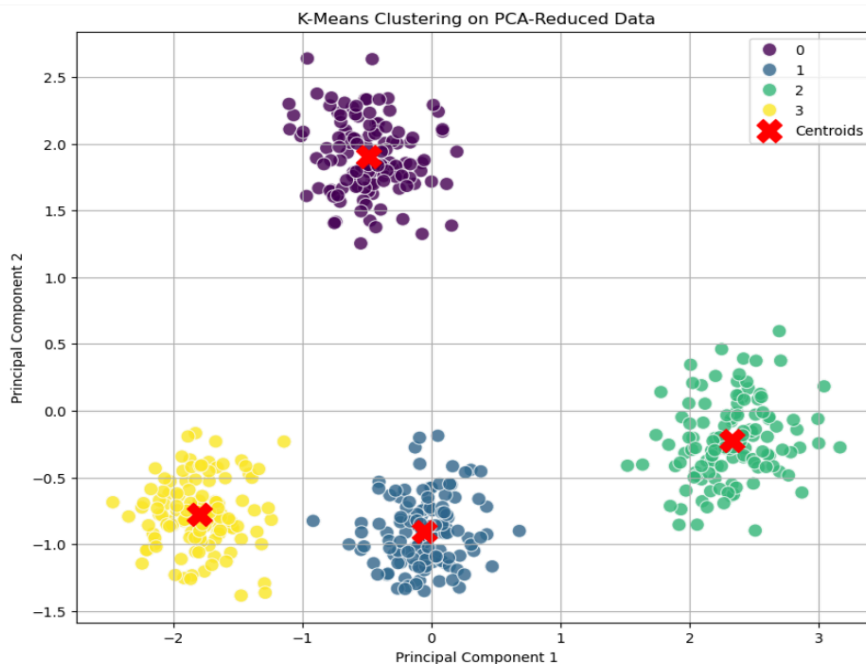
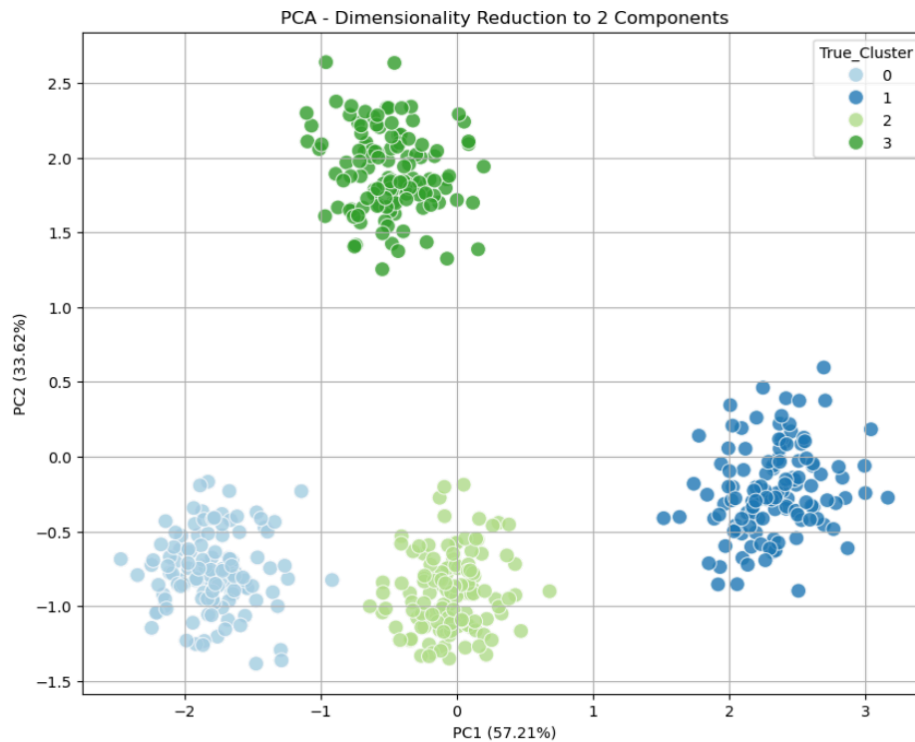
Original PCA Dataset Shape: (500, 5)

Principal Components Head:

	Principal_Component_1	Principal_Component_2	True_Cluster
0	-0.455305	1.623917	3
1	2.705622	0.375012	1
2	-0.810234	1.966926	3
3	-0.427139	2.149626	3
4	2.407508	0.099250	1

Explained Variance Ratio: [0.57208431 0.33622342]

Total Explained Variance by 2 PCs: 0.908



Silhouette Score for K-Means on PCA (K=4): 0.776

RESULT:

K-Means and PCA were successfully implemented. K-Means grouped the data into clusters, while PCA reduced the dimensions and enabled effective visualization of the data.